```
!pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
```

Double-click (or enter) to edit

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.5)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files sales_analysis.csv

sales_analysis.csv(text/csv) - 381 bytes, last modified: 10/26/2026 - 100% done

```
Saving sales_analysis.csv to sales_analysis (1).csv
```

```
import numpy as np
import pandas as pd
df = pd.read_csv('sales_analysis.csv')
print("File loaded Successfully!")
df.head()
```

```
File loaded Successfully!
```

	Order	Category	City	Sales	Quantity
0	101	Electronics	Chennai	5000	2
1	102	Clothing	Bangalore	2000	4
2	103	Electronics	Chennai	7000	3
3	104	Food	Hyderabad	1500	5
4	105	Clothing	Bangalore	3000	2

Next steps: Generate code with df | New interactive sheet

```
sales_array = np.array(df["Sales"])
print("Sales array:", sales_array)
print("Minimum sales", np.min(sales_array))
print("Maximum sales", np.max(sales_array))
print("Mean sales", np.mean(sales_array))
print("Median sales", np.median(sales_array))
print("Standard deviation of sales", np.std(sales_array))
```

```
Sales array: [5000 2000 7000 1500 3000 2500 6000 4000 1800 8500 3200 2200]
Minimum sales 1500
Maximum sales 8500
Mean sales 3891.6666666666665
Median sales 3100.0
Standard deviation of sales 2166.5865369798234
```

```
df.head()
```

	Order	Category	City	Sales	Quantity
0	101	Electronics	Chennai	5000	2
1	102	Clothing	Bangalore	2000	4
2	103	Electronics	Chennai	7000	3
3	104	Food	Hyderabad	1500	5
4	105	Clothing	Bangalore	3000	2

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	Order	Category	City	Sales	Quantity
7	108	Clothing	Chennai	4000	3
8	109	Food	Bangalore	1800	4
9	110	Electronics	Chennai	8500	1
10	111	Clothing	Hyderabad	3200	2
11	112	Food	Bangalore	2200	5

```
print("Shape:", df.shape)
print("Size:", df.size)
print("Dimensions:", df.ndim)
print("Column names:",df.columns)
print("Number of rows:",df.shape[0])
print("Number of columns:",df.shape[1])
```

```
Shape: (12, 5)
Size: 60
Dimensions: 2
Column names: Index(['Order', 'Category', 'City', 'Sales', 'Quantity'], dtype='object')
Number of rows: 12
Number of columns: 5
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Order     12 non-null     int64
 1   Category  12 non-null     object
 2   City      12 non-null     object
 3   Sales     12 non-null     int64
 4   Quantity  12 non-null     int64
dtypes: int64(3), object(2)
memory usage: 612.0+ bytes
```

```
df.describe()
```

	Order	Sales	Quantity
count	12.000000	12.000000	12.000000
mean	106.500000	3891.666667	3.250000
std	3.605551	2262.925835	1.544786
min	101.000000	1500.000000	1.000000
25%	103.750000	2150.000000	2.000000
50%	106.500000	3100.000000	3.000000
75%	109.250000	5250.000000	4.250000
max	112.000000	8500.000000	6.000000

```
df.isnull().sum()
```

	0
Order	0
Category	0
City	0
Sales	0
Quantity	0

dtype: int64

```
print("Number of duplicated rows:",df.duplicated().sum())
```

```
print("Number of duplicated rows:",df.duplicated().sum())
```

```
Number of duplicated rows: 0
```

```
print("Minimum sales:", df["Sales"].min())
print("Maximum sales:", df["Sales"].max())
print("Average sales:", df["Sales"].mean())
print("Total sales:", df["Sales"].sum())
```

```
Minimum sales: 1500
Maximum sales: 8500
Average sales: 3891.6666666666665
Total sales: 46700
```

```
print(df.groupby("Category")["Sales"].sum())
```

```
Category
Clothing       12200
Electronics    26500
Food            8000
Name: Sales, dtype: int64
```

```
print(df[["Order", "Sales"]])
```

```
    Order  Sales
0     101   5000
1     102   2000
2     103   7000
3     104   1500
4     105   3000
5     106   2500
6     107   6000
7     108   4000
8     109   1800
9     110   8500
10    111   3200
11    112   2200
```

```
print(df.groupby("City")["Sales"].sum())
```

```
City
Bangalore     9000
Chennai      27000
Hyderabad    10700
Name: Sales, dtype: int64
```

```
df[df["City"] == "Chennai"]
```

	Order	Category	City	Sales	Quantity
0	101	Electronics	Chennai	5000	2
2	103	Electronics	Chennai	7000	3
5	106	Food	Chennai	2500	6
7	108	Clothing	Chennai	4000	3
9	110	Electronics	Chennai	8500	1

```
df[df["Category"] == "Electronics"]
```

	Order	Category	City	Sales	Quantity
0	101	Electronics	Chennai	5000	2
2	103	Electronics	Chennai	7000	3
6	107	Electronics	Hyderabad	6000	2
9	110	Electronics	Chennai	8500	1

```
df[(df["Category"] == "Electronics") & (df["City"] == "Chennai")]
```

	Order	Category	City	Sales	Quantity
0	101	Electronics	Chennai	5000	2
2	103	Electronics	Chennai	7000	3
9	110	Electronics	Chennai	8500	1

```
df.groupby(["Category", "City"])["Sales"].sum().reset_index()
```

	Category	City	Sales
0	Clothing	Bangalore	5000
1	Clothing	Chennai	4000
2	Clothing	Hyderabad	3200
3	Electronics	Chennai	20500
4	Electronics	Hyderabad	6000
5	Food	Bangalore	4000
6	Food	Chennai	2500
7	Food	Hyderabad	1500